



Session 11A: Trigonometry Foundations — Angles, Circles, and Waves

Phase 2 — Classical Techniques | 75 min

Part A: Angles — From Degrees to Radians

Example 1: Remember Only $\pi = 180^\circ$

Radian measure: an angle that cuts an arc equal to the radius on the unit circle = 1 radian.

A semicircle has arc length $\pi r \rightarrow 180^\circ = \pi \text{ rad}$.

Convert using proportion:

$$90^\circ = \frac{\pi}{2}, 60^\circ = \frac{\pi}{3}, 45^\circ = \frac{\pi}{4}, 30^\circ = \frac{\pi}{6}.$$

$$120^\circ = \frac{2\pi}{3}, 270^\circ = \frac{3\pi}{2}, 360^\circ = 2\pi.$$

$$\text{Reverse: } \frac{5\pi}{6} \times \frac{180^\circ}{\pi} = 150^\circ.$$

$$\frac{7\pi}{4} = 315^\circ, \frac{11\pi}{6} = 330^\circ.$$

Part B: The Unit Circle — The Seed of All Trigonometry

Example 2: Read $(\cos \theta, \sin \theta)$ from the Unit Circle

On a circle of radius 1, go around by angle θ . The coordinates of the point reached = $(\cos \theta, \sin \theta)$.

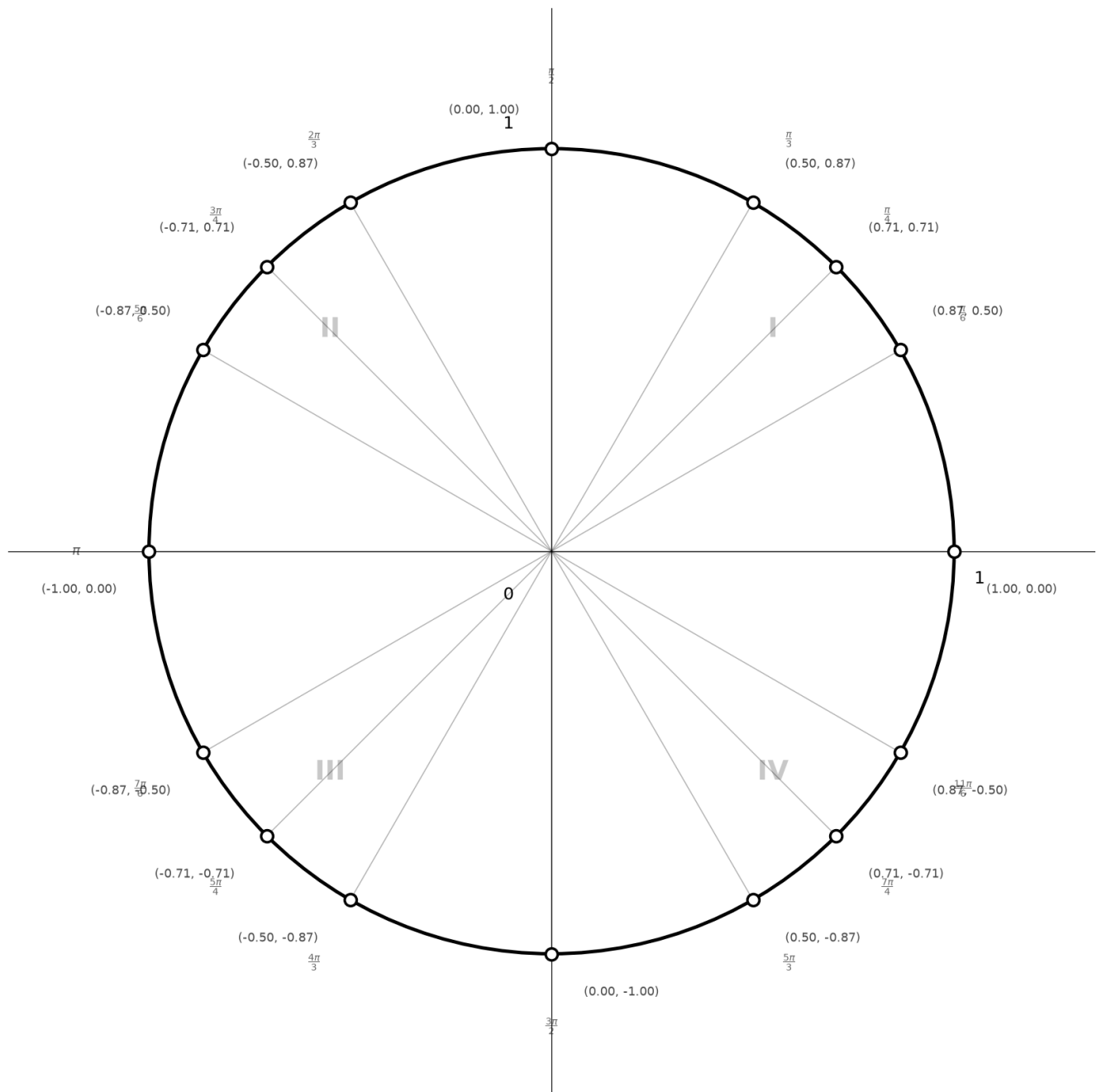
$$\theta = 0: (1, 0). \theta = \frac{\pi}{2}: (0, 1). \theta = \pi: (-1, 0). \theta = \frac{3\pi}{2}: (0, -1).$$

$$\theta = \frac{\pi}{4}: x^2 + y^2 = 1, x = y \rightarrow x = y = \frac{\sqrt{2}}{2}. \rightarrow \cos \frac{\pi}{4} = \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}.$$

$$\theta = \frac{\pi}{3}: 30\text{-}60\text{-}90 \text{ triangle. Short leg } \frac{1}{2}, \text{ long leg } \frac{\sqrt{3}}{2}.$$

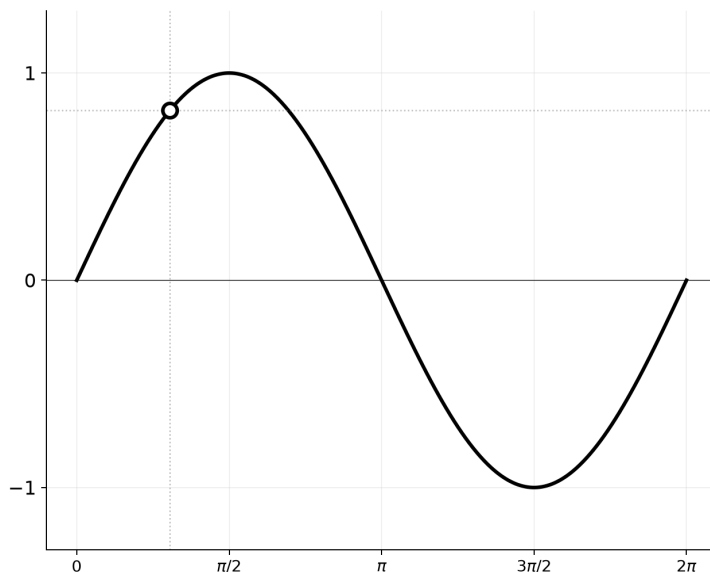
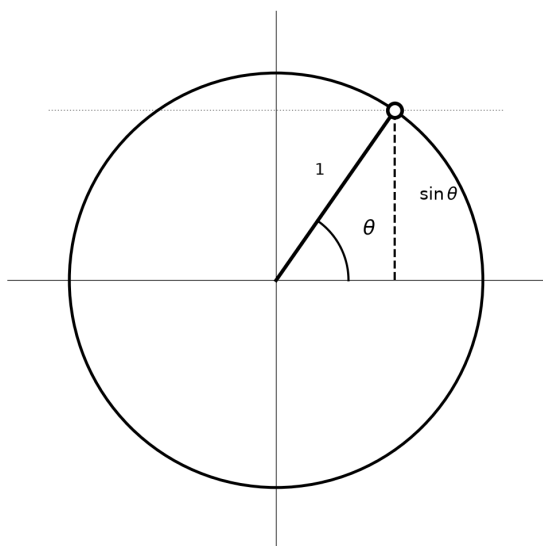
$$\rightarrow \cos \frac{\pi}{3} = \frac{1}{2}, \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}.$$

$$\theta = \frac{\pi}{6}: \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}, \sin \frac{\pi}{6} = \frac{1}{2}.$$



Visual Interlude: Unwrapping the Circle into the Sine Wave

A point moves counterclockwise around the unit circle at constant speed. Below it, a graph tracks its height (y -coordinate) against angle θ .



At $\theta = 0$: height 0 $\rightarrow$ sine starts at 0. At $\theta = \frac{\pi}{2}$: height 1 $\rightarrow$ peak. At $\theta = \pi$: height 0 $\rightarrow$ crosses axis. At $\theta = \frac{3\pi}{2}$: height -1 $\rightarrow$ trough. At $\theta = 2\pi$: back to 0 — one full cycle.

Cosine unwraps the x -coordinate instead — starts at 1, shifted left by $\frac{\pi}{2}$.

Tangent unwrapped: draw a vertical line at $(1, 0)$. Extend the radius — where it hits the tangent line, the y -coordinate is $\tan \theta$. As $\theta \rightarrow 90^\circ$, the radius parallels the tangent line (they never meet) $\rightarrow \tan \theta \rightarrow \infty$.

Example 3: Engrave the Special Angle Table in Your Hand

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined

sin: $0, \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, 1$ — increasing.

cos: same sequence in reverse. $\tan = \frac{\sin}{\cos}$.

Example 4: Quadrant Signs — "All Students Take Calculus"

$\tan \theta = \frac{\sin \theta}{\cos \theta}$. Reciprocal functions follow sign of their originals.

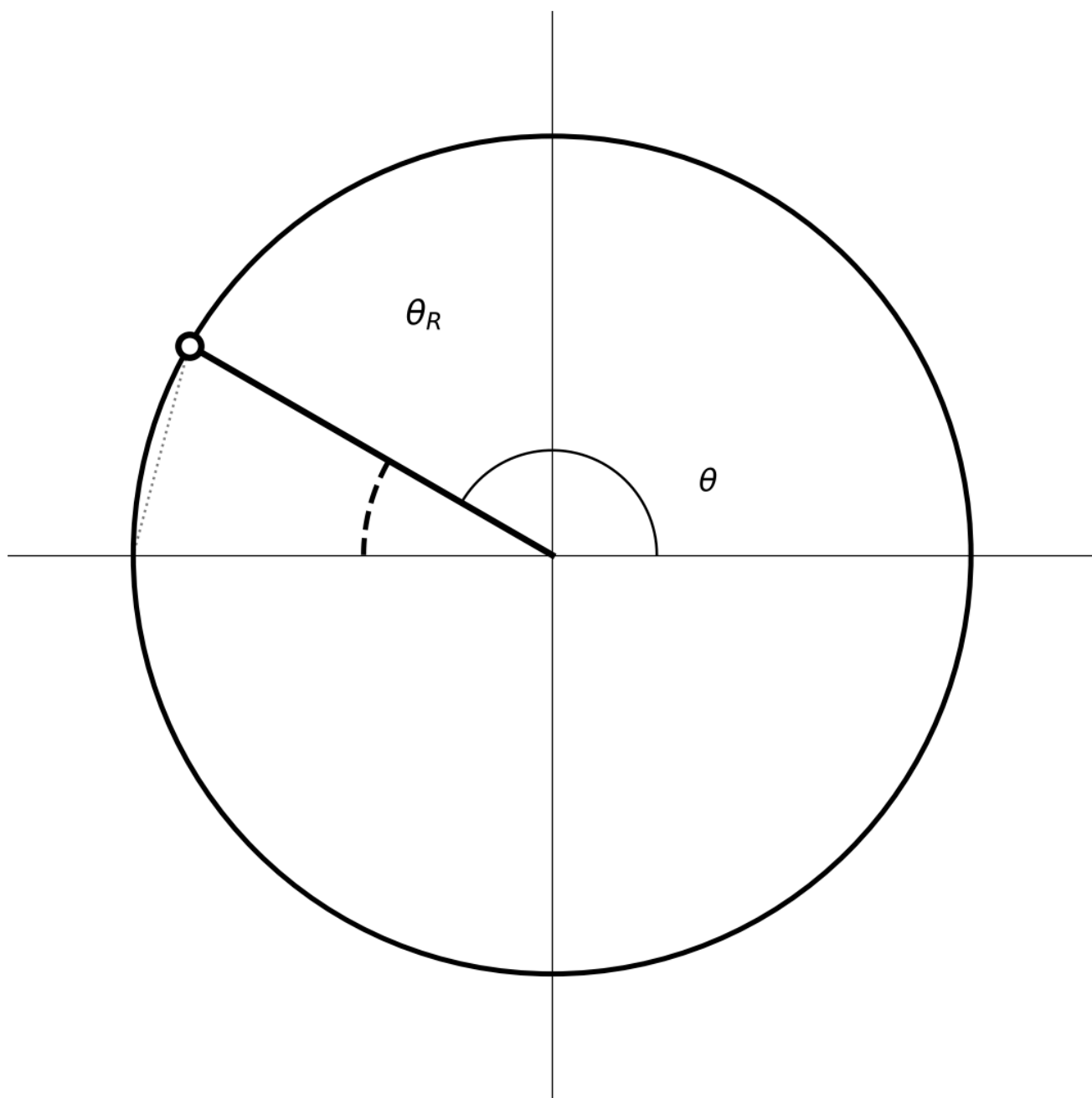
Quadrant	sin	cos	tan	csc	sec	cot
1	+	+	+	+	+	+
2	+	−	−	+	−	−
3	−	−	+	−	−	+
4	−	+	−	−	+	−

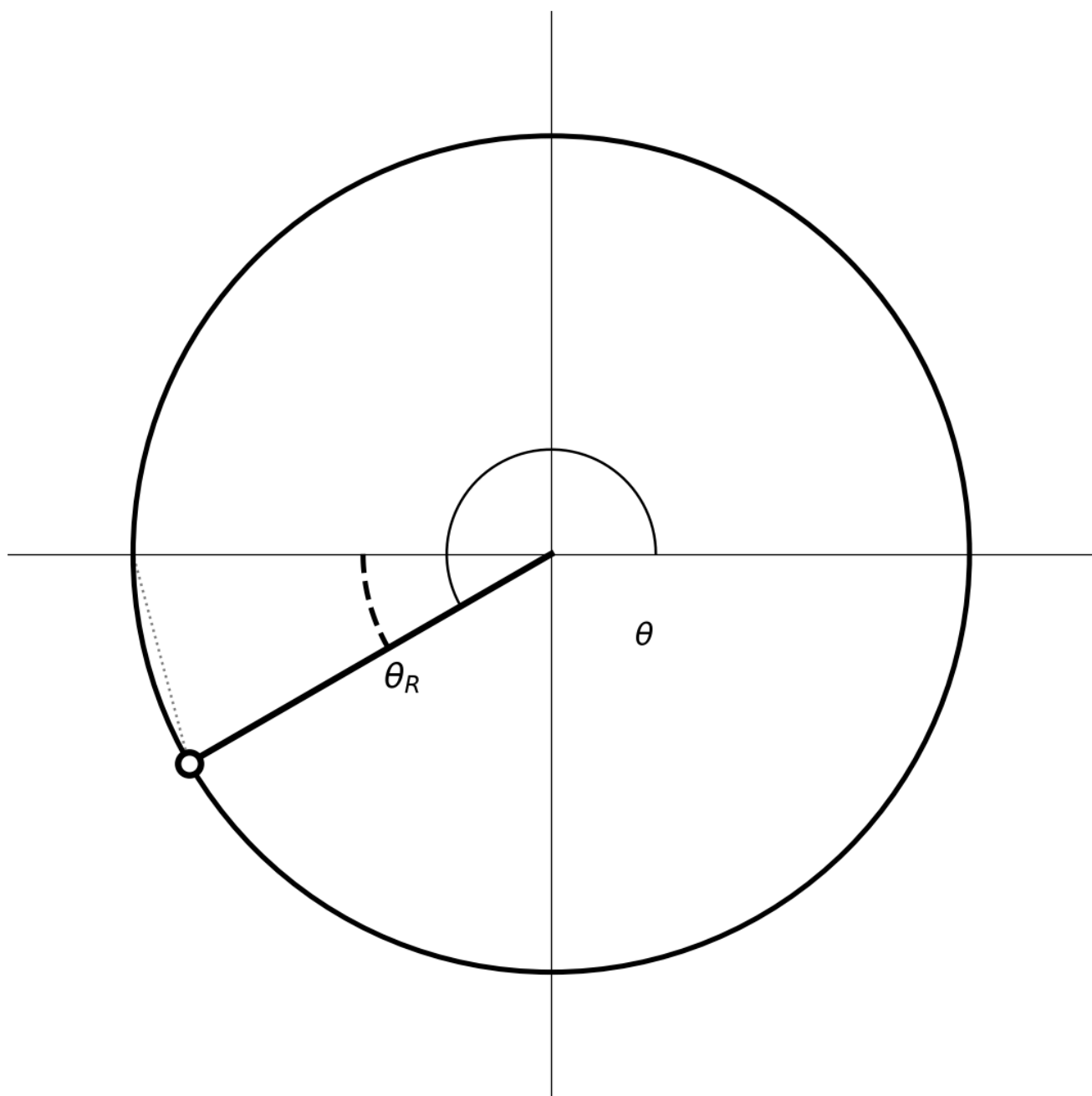
$\sin 150^\circ$: Q2 $\rightarrow$ +. $150^\circ = 180^\circ - 30^\circ \rightarrow \sin 30^\circ = \frac{1}{2}$.
 $\cos 210^\circ$: Q3 $\rightarrow$ −. $210^\circ = 180^\circ + 30^\circ \rightarrow -\cos 30^\circ = -\frac{\sqrt{3}}{2}$.
 $\tan 300^\circ$: Q4 $\rightarrow$ −. $300^\circ = 360^\circ - 60^\circ \rightarrow -\tan 60^\circ = -\sqrt{3}$.

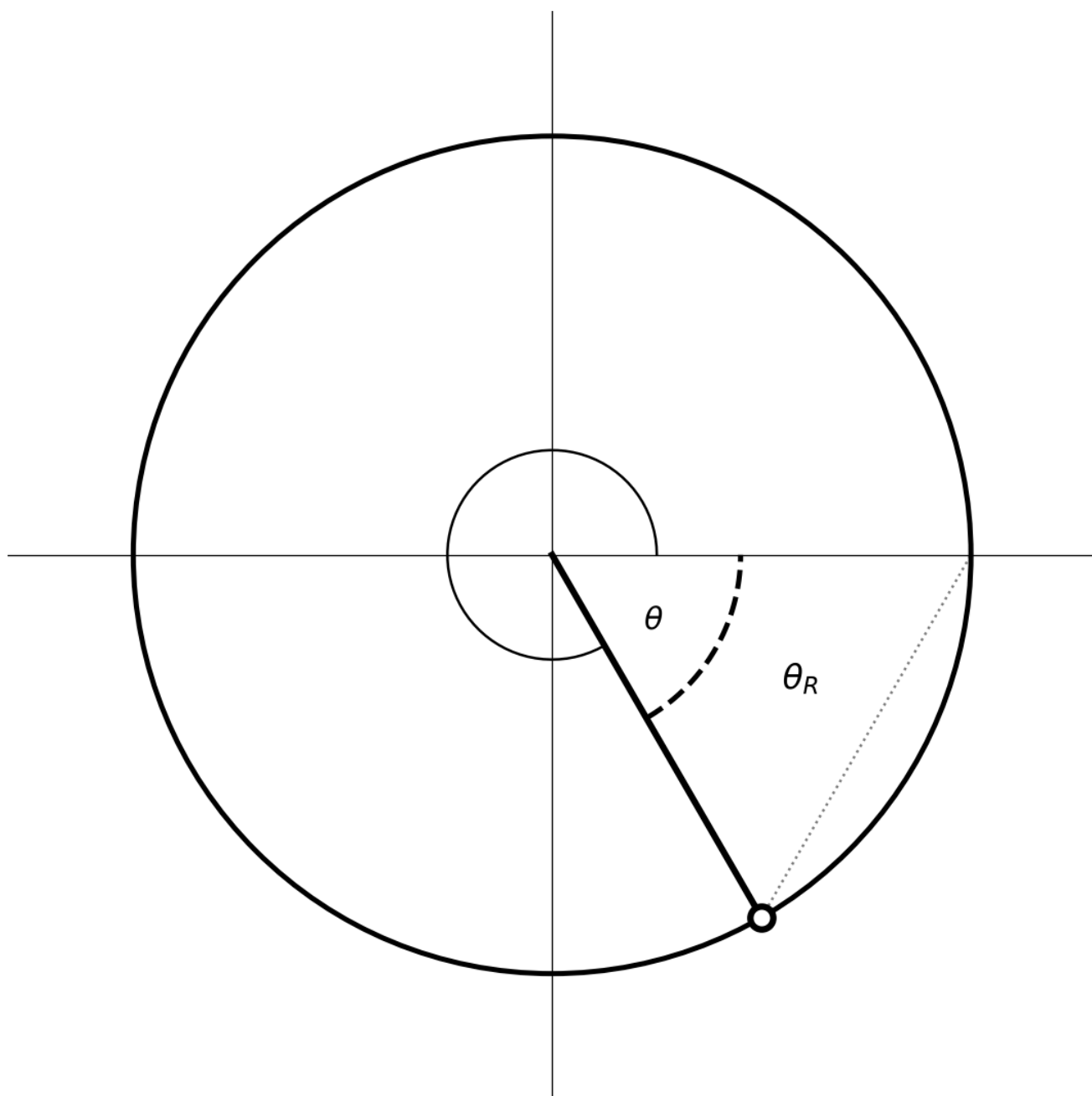
Reference angle method: Find the acute angle to the nearest x -axis. Apply quadrant sign.

Visual Interlude: The Reference Angle — A Geometric Shortcut

Every angle θ has a "shadow" in Quadrant 1 — its reference angle θ_R , the acute angle between the terminal side and the x -axis.







The rule: The *magnitude* of any trig value at any angle equals its value at the reference angle. Only the sign changes — determined by the quadrant.

Example 5: csc, sec, cot — Completing the Family of Six

$$\csc \theta = \frac{1}{\sin \theta} \ (\sin \theta \neq 0). \sec \theta = \frac{1}{\cos \theta} \ (\cos \theta \neq 0).$$
$$\cot \theta = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta} \ (\sin \theta \neq 0).$$

Evaluate:

$$\csc \frac{\pi}{6} = \frac{1}{1/2} = 2. \sec \frac{\pi}{4} = \frac{1}{\sqrt{2}/2} = \sqrt{2}.$$

$$\cot \frac{\pi}{3} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}.$$

$$\csc \frac{5\pi}{6} = \frac{1}{\sin 150^\circ} = \frac{1}{1/2} = 2.$$

Extended Pythagorean identities:

$$\sin^2 \theta + \cos^2 \theta = 1.$$

$$1 + \tan^2 \theta = \sec^2 \theta \ (\text{divide the above by } \cos^2 \theta).$$

$$1 + \cot^2 \theta = \csc^2 \theta \ (\text{divide the above by } \sin^2 \theta).$$

Part C: Graphs — Drawing and Shaping Waves

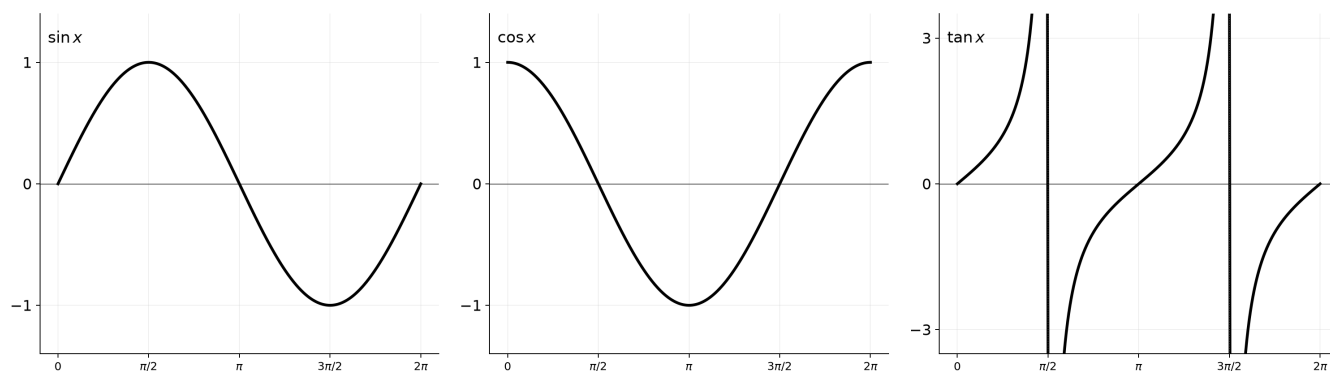
Example 6: Basic Graphs of $\sin$, $\cos$, $\tan$

$y = \sin x$: starts at $(0, 0)$, period 2π , amplitude 1, range $[-1, 1]$.

Hits 1 at $\frac{\pi}{2}$, 0 at π , -1 at $\frac{3\pi}{2}$, 0 at 2π . Origin-symmetric (odd).

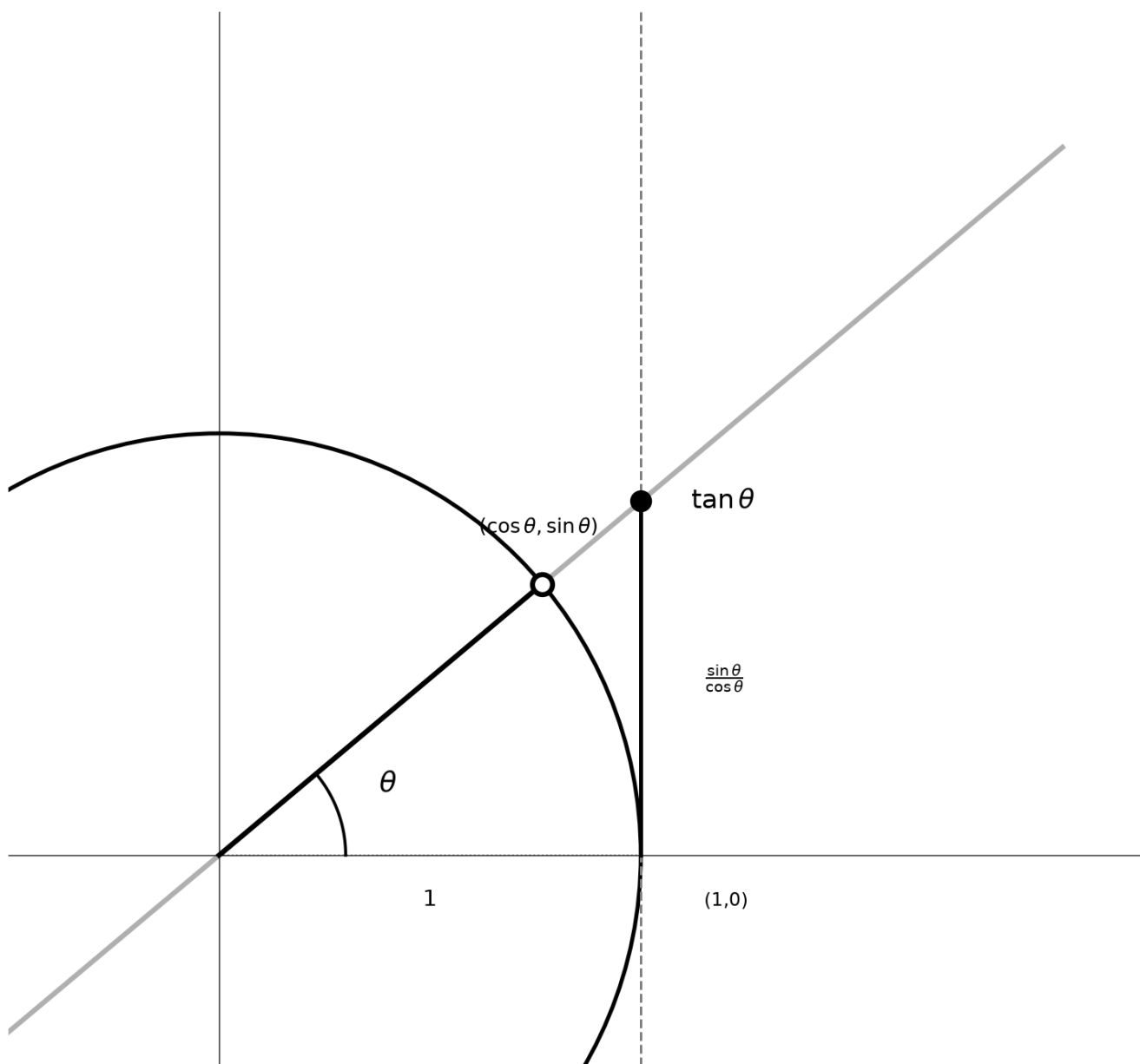
$y = \cos x$: starts at $(0, 1)$. $\sin$ shifted left by $\frac{\pi}{2}$. y -axis-symmetric (even).

$y = \tan x$: vertical asymptotes where $\cos = 0$ ($\frac{\pi}{2}, \frac{3\pi}{2}, \dots$). Period π . Origin-symmetric (odd).



Visual Insight — Tangent as Slope:

On the unit circle, the radius to $(\cos \theta, \sin \theta)$ has slope $\frac{\sin \theta}{\cos \theta} = \tan \theta$.



Visual Insight — $\sin \theta$ and $\cos \theta$ as Projections:

Drop a perpendicular from the circle-point to the x -axis: the x -intercept is $\cos \theta$, the y -intercept is $\sin \theta$. They are the horizontal and vertical shadows of a rotating unit vector.

Example 7: Graphs of $\csc$, $\sec$, $\cot$

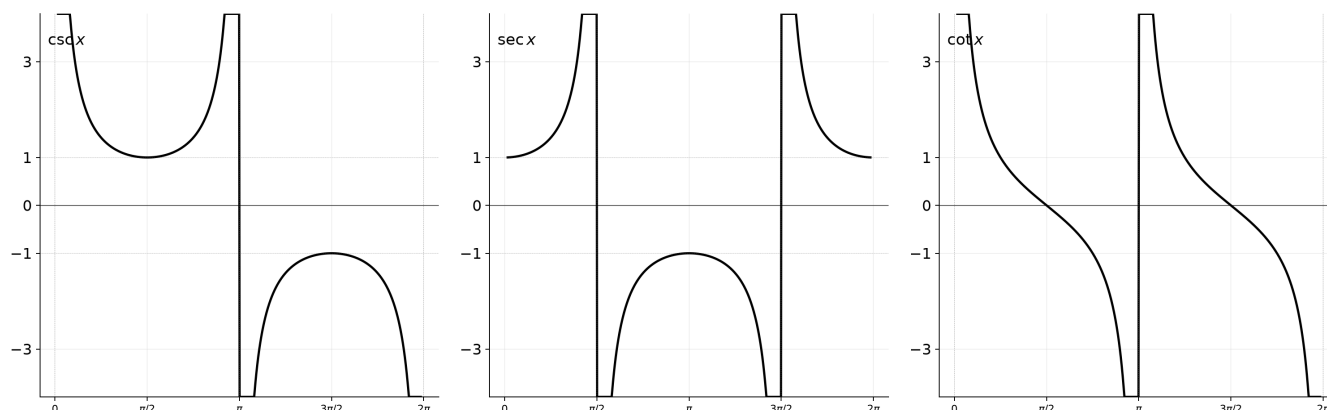
$\csc x = \frac{1}{\sin x}$: asymptotes at $0, \pi, 2\pi, \dots$ (where $\sin = 0$).

U-shaped branches above $y = 1$ where $\sin > 0$; $\cap$ -shaped below $y = -1$ where $\sin < 0$.

$\sec x = \frac{1}{\cos x}$: asymptotes at $\frac{\pi}{2}, \frac{3\pi}{2}, \dots$ (where $\cos = 0$).

Branches above $y = 1$ and below $y = -1$.

$\cot x = \frac{1}{\tan x} = \frac{\cos}{\sin}$: asymptotes at $0, \pi, 2\pi, \dots$ (where $\sin = 0$). Period π , always decreasing.



Example 8: $y = a \sin(bx + c) + d$ — Cooking a Wave

$$y = 3 \sin\left(2x - \frac{\pi}{3}\right) + 1.$$

(1) **Amplitude:** $|a| = 3$. Wave height ± 3 from center.

(2) **Period:** $\frac{2\pi}{|b|} = \frac{2\pi}{2} = \pi$.

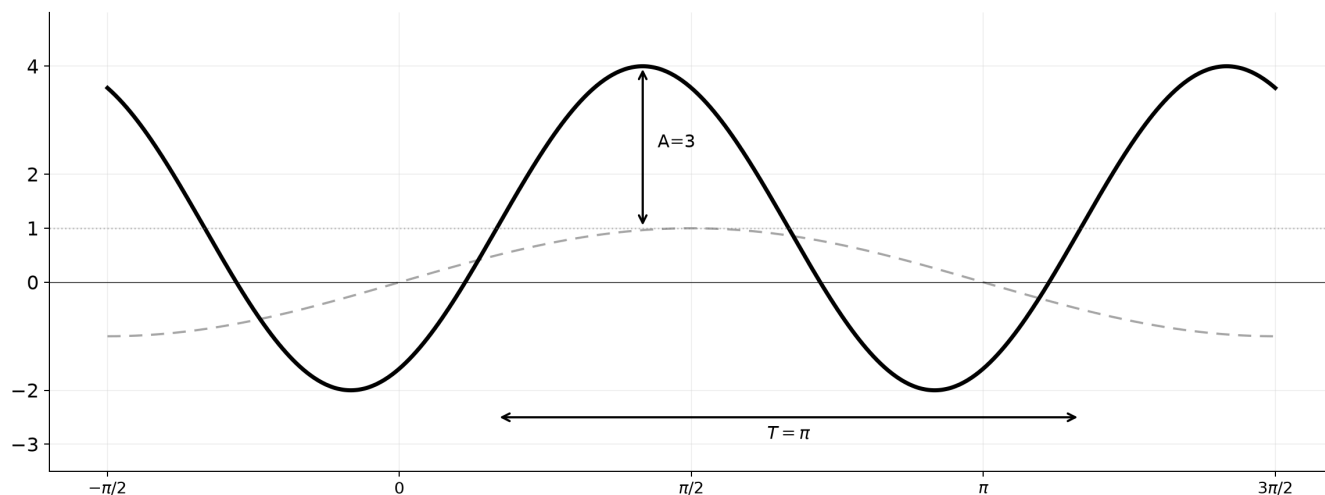
(3) **Phase shift:** $bx + c = 0 \rightarrow x = -\frac{c}{b} = \frac{\pi}{6}$. Right by $\frac{\pi}{6}$.

(4) **Vertical shift:** $+1$. Range $[-2, 4]$.

$y = -2 \cos\left(\frac{x}{2}\right)$: amplitude 2 (flipped), period 4π .

$y = 4 \sin\left(\pi x + \frac{\pi}{4}\right) - 3$:

Amplitude 4, period $\frac{2\pi}{\pi} = 2$, phase $x = -\frac{1}{4}$ (left by $\frac{1}{4}$), range $[-7, 1]$.

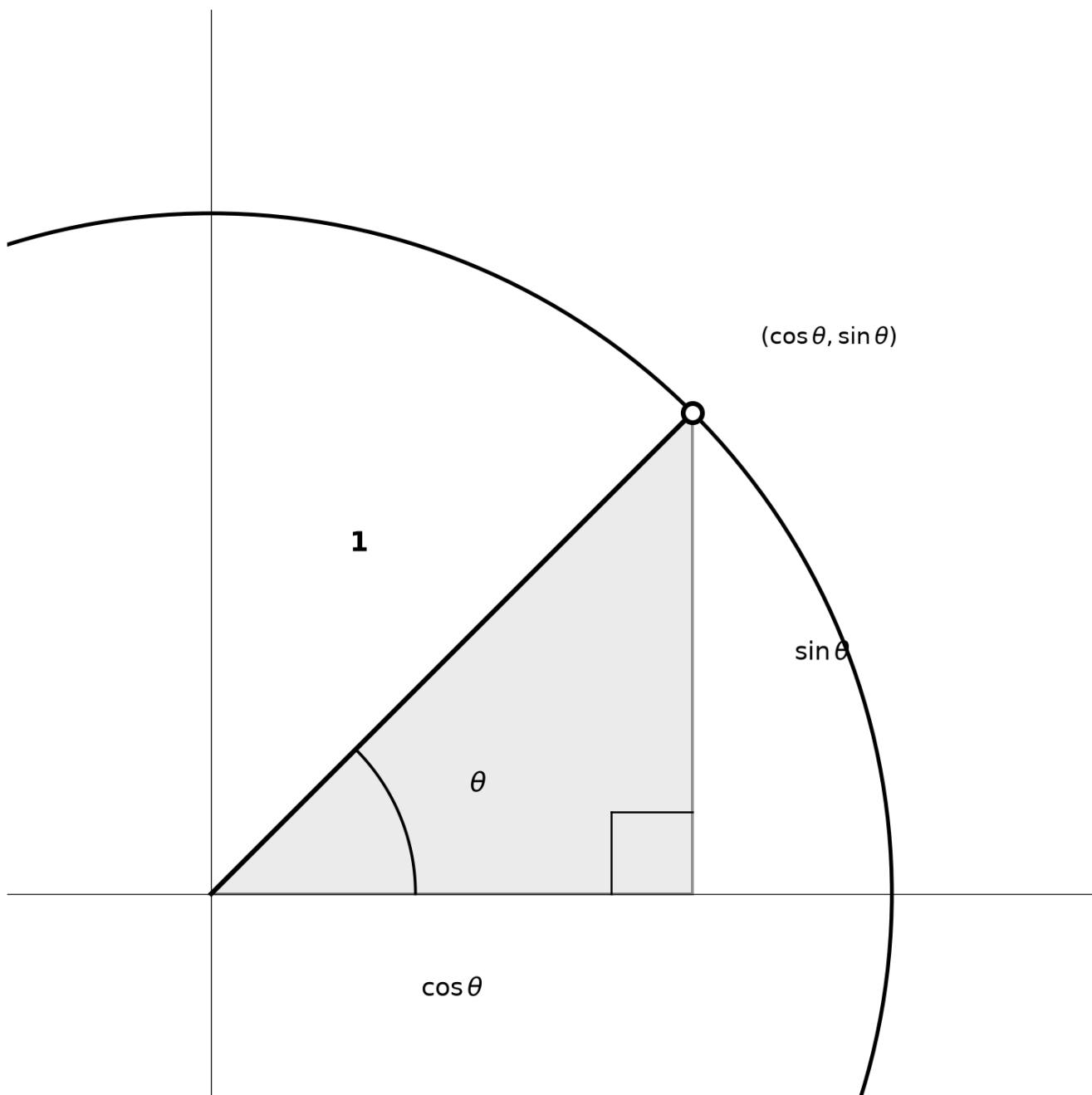


Up to here: Radians from $\pi = 180^\circ$. Unit circle $\rightarrow (\cos \theta, \sin \theta)$. Special angle table engraved.

6 trig functions + quadrant signs. Graphs of all 6 functions. Wave shaping with a, b, c, d .

Visual Interlude: Geometric Proof of $\sin^2 \theta + \cos^2 \theta = 1$

The point $(\cos \theta, \sin \theta)$ lies on the circle $x^2 + y^2 = 1$. Substitute: $\cos^2 \theta + \sin^2 \theta = 1$. This is the Pythagorean theorem — no algebra needed.



From this one picture flow all the Pythagorean identities: divide by $\cos^2 \theta \rightarrow 1 + \tan^2 \theta = \sec^2 \theta$
; divide by $\sin^2 \theta \rightarrow 1 + \cot^2 \theta = \csc^2 \theta$.

Common Mistakes

Mistake 1: $\sin 2x = 2 \sin x$

Wrong path: " $\sin 2x = 2 \sin x$."

Why wrong: $\sin 2x = 2 \sin x \cos x$. The factor $\cos x$ is missing.

Right path: $\sin 2\theta = 2 \sin \theta \cos \theta$. Always write both factors.

Mistake 2: Forgetting arcsin range

Wrong path: " $\sin x = \frac{1}{2} \rightarrow x = \arcsin \frac{1}{2} = \frac{\pi}{6}$ or $\frac{5\pi}{6}$."

Why wrong: $\arcsin$ returns values only in $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

Right path: $\arcsin \frac{1}{2} = \frac{\pi}{6}$. Find the second solution manually: $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$.

Mistake 3: Confusing period of $\tan x$

Wrong path: "The period of $\tan x$ is 2π , same as $\sin$ and $\cos$."

Why wrong: $\tan x = \frac{\sin x}{\cos x}$ repeats every π because both numerator and denominator flip sign after π , canceling out.

Right path: Period of $\tan x$ is π . Asymptotes occur every π as well.

What We Just Did

- (1) Radian measure – $\pi = 180^\circ$, convert by proportion. Memorize the common angles.
The unit circle: a point at angle θ has coordinates $(\cos \theta, \sin \theta)$.
Read sine and cosine directly from the circle. Special angle table engraved.
- (2) Six functions – $\sin, \cos, \tan$ plus their reciprocals $\csc, \sec, \cot$.
Quadrant signs: All positive in Q1, Sin positive in Q2, Tan in Q3, Cos in Q4.
Reference angle: reduce any angle to its Q1 acute shadow.
- (3) Graphs – $\sin$ starts at 0, $\cos$ starts at 1, $\tan$ has asymptotes.
 $y = a \cdot \sin(bx+c)+d$: amplitude= $|a|$, period= $2\pi/|b|$, phase= $-c/b$, v-shift= d .
Tangent is the slope of the radius. $\sin^2+\cos^2=1$ is the Pythagorean theorem.

Practice 1

$\sin \theta = -\frac{\sqrt{3}}{2}$, θ in Q4. Find $\cos \theta$, $\tan \theta$, $\sec \theta$, $\csc \theta$, $\cot \theta$.

→ Reference: **Example 3, 4, 5**

Solutions: [Solutions](#)

Practice 2

For $y = 2 \sin(3x + \pi) - 1$, give the amplitude, period, phase shift, and vertical shift. State the max and min values.

→ Reference: **Example 8**

Solutions: [Solutions](#)

Practice 3: Composition

Draw a rough sketch of one period of $y = -3 \cos(2x) + 2$. Label the maximum, minimum, and axis intercepts. Then explain in words how each parameter (a , b , d) changed the graph from the basic $y = \cos x$.

→ Reference: **Example 6, 8**

Solutions: [Solutions](#)

Practice 4

Convert to radians and evaluate exactly: $\sin 210^\circ$, $\cos 315^\circ$, $\tan 240^\circ$.

→ Reference: **Example 1, 4**

Solutions: [Solutions](#)

Practice 5: Real Battle

A Ferris wheel has radius 30 m and completes one revolution in 4 minutes. The lowest point is 2 m above the ground. Write a function $h(t) = a \sin(bt + c) + d$ for the height above ground after t minutes, assuming you start at the lowest point.

→ Reference: **Example 8**

Solutions: [Solutions](#)

Basic Algebra Drill — Trigonometry Foundations (10 Problems)

Pure calculation. Every fundamental operation appears at least once. Build speed.

D1. Convert to radians: 150° , 225° , 330° .

D2. Convert to degrees: $\frac{3\pi}{4}$, $\frac{5\pi}{3}$, $\frac{11\pi}{6}$.

D3. Evaluate without a calculator: $\sin \frac{5\pi}{6}$, $\cos \frac{7\pi}{4}$, $\tan \frac{4\pi}{3}$.

D4. Evaluate without a calculator: $\sin \frac{3\pi}{2}$, $\cos \pi$, $\tan \frac{5\pi}{4}$.

D5. Given $\sin \theta = \frac{5}{13}$ and θ in Q2, find $\cos \theta$ and $\tan \theta$.

D6. Given $\cos \theta = -\frac{3}{5}$ and θ in Q3, find $\sin \theta$, $\tan \theta$, $\sec \theta$, $\csc \theta$, $\cot \theta$.

D7. Find the period and amplitude of $y = -4 \cos\left(\frac{\pi}{2}x\right) + 3$.

D8. Find the amplitude, period, phase shift, and vertical shift of $y = 5 \sin(4x - \pi) - 2$.

D9. Simplify $\frac{\sin x}{\csc x} + \frac{\cos x}{\sec x}$. (Write $\csc$ and $\sec$ in terms of $\sin$ and $\cos$ first.)

D10. Evaluate $\sin^2 15^\circ + \sin^2 75^\circ$. (Hint: $\sin 75^\circ = \cos 15^\circ$.)

Solutions: [Solutions](#)

Advanced Algebra Drill — Trigonometry Foundations (10 Problems)

Multi-step. Moves beyond single-step evaluation into identity chaining and equation solving.

A1. Find all six trigonometric functions of $\theta = \frac{5\pi}{3}$.

A2. If $\sin \theta = \frac{4}{5}$ and $\theta \in Q1$, compute $\sin 2\theta$ and $\cos 2\theta$ without finding θ first.

A3. Simplify $\frac{\sin(-x)}{\cos(-x)} + \frac{\cos(-x)}{\sin(-x)}$. Use odd/even properties.

- A4.** Evaluate $\sin \frac{2\pi}{3} \cdot \cos \frac{\pi}{6} + \cos \frac{2\pi}{3} \cdot \sin \frac{\pi}{6}$ by recognizing the sum formula pattern.
- A5.** Write $y = -2 \sin(3x + \frac{\pi}{2}) + 1$ in the form $y = a \cos(bx + c) + d$. (Hint: $\sin(\theta + \frac{\pi}{2}) = \cos \theta$.)
- A6.** Find all $x \in [0, 2\pi]$ such that $2 \sin^2 x - 1 = 0$.
- A7.** If $\tan \theta = 2$ and $\theta \in Q3$, find $\sin \theta$ and $\cos \theta$ exactly. (Draw the triangle, apply quadrant signs.)
- A8.** Simplify $\frac{1-\cos^2 \theta}{\sin \theta} + \frac{1-\sin^2 \theta}{\cos \theta}$ to a single trig function.
- A9.** A wave has equation $y = 3 \cos(\frac{\pi}{4}x) - 1$. Find all $x \in [0, 16]$ where $y = -1$.
- A10.** Prove that $\sin^4 \theta + \cos^4 \theta = 1 - \frac{1}{2} \sin^2 2\theta$.

Solutions: [Solutions](#)

Today's Procedure

- Step 1: Radians – $\pi = 180^\circ$. Convert by multiplying by $\pi/180^\circ$ (to radians) or $180^\circ/\pi$ (to degrees). Memorize the 5 special angles: $0, \pi/6, \pi/4, \pi/3, \pi/2$. Read $(\cos \theta, \sin \theta)$ directly from the unit circle.
- Step 2: Six functions – $\sin, \cos, \tan, \csc=1/\sin, \sec=1/\cos, \cot=1/\tan$.
 Quadrant signs: All-Sin-Tan-Cos. Reference angle = acute angle to x-axis.
 Pythagorean identity: $\sin^2+\cos^2=1$, plus its two variants.
- Step 3: Graphs – $\sin$: starts at 0, period 2π . $\cos$: starts at 1, period 2π .
 $\tan$: asymptotes at $\pi/2 + n\pi$, period π .
 $y = a \cdot \sin(bx+c)+d$: a = amplitude, $2\pi/|b|$ = period, $-c/b$ = phase, d = v-shift.

How to Read These Symbols

Symbol	Reads as	Meaning
$\sin \theta$	"sine theta"	opposite / hypotenuse in right triangle; y-coordinate on unit circle
$\cos \theta$	"cosine theta"	adjacent / hypotenuse; x-coordinate on unit circle
$\tan \theta$	"tangent theta"	$\sin/\cos$ = opposite/adjacent; slope of terminal ray
$\csc \theta$	"cosecant theta"	$1/\sin$ — reciprocal of sine
$\sec \theta$	"secant theta"	$1/\cos$ — reciprocal of cosine
$\cot \theta$	"cotangent theta"	$1/\tan = \cos/\sin$ — reciprocal of tangent
$\pi \text{ rad} = 180^\circ$	"pi radians equals 180 degrees"	radian-degree conversion
$\sin^2 \theta + \cos^2 \theta = 1$	"sine squared theta plus cosine squared theta equals one"	Pythagorean identity
$\sin(2\theta) = 2 \sin \theta \cos \theta$	"sine two theta equals two sine theta cosine theta"	double-angle formula for sine
$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$	"cosine two theta equals cos squared theta minus sin squared theta"	double-angle formula for cosine
2π	"two pi" / "tau"	period of sine and cosine (one full circle)
π	"pi"	period of tangent
$f(x) = A \sin(Bx + C) + D$	"A sine B x plus C plus D"	sinusoid: A=amplitude, $2\pi/B$ =period, $-C/B$ =phase shift, D=vertical shift

Terminology

Up to now we used plain words like "radian", "unit circle", "wave", "one full turn".

You have already learned all the methods. Now we attach the formal mathematical names.

What we called it	Mathematical term	Notation
radian	radian	$\pi = 180^\circ$
unit circle	unit circle	$x^2 + y^2 = 1$
period / one full turn	period	sin/cos: 2π , tan: π
amplitude / wave height	amplitude	a
phase shift	phase shift	$-c/b$
reciprocal trig functions	reciprocal trigonometric functions	csc, sec, cot
Pythagorean identity	Pythagorean identity	$\sin^2 \theta + \cos^2 \theta = 1$
reference angle	reference angle	acute angle to nearest x -axis